

AI HOME DESIGNER
REST API SPECIFICATION
PART 1 + PART 2

STATUS:
API FOUNDATION + AUTHENTICATION

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1. API OVERVIEW

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ARCHITECTURE

REST API

JSON

HTTPS ONLY

UTF-8

BASE URL

/api/v1

AUTHENTICATION

JWT Access Token

Refresh Token

=====

=====

2. HTTP METHODS

=====

=====

GET

Read

POST

Create

PUT

Replace

PATCH

Update

DELETE

Remove

=====

3. STANDARD RESPONSE

=====

SUCCESS

```
{
  "success": true,
  "data": {},
  "meta": {}
}
```

ERROR

```
{
  "success": false,
  "error": {
    "code": "...",
    "message": "...",
    "details": []
  }
}
```

=====

4. AUTH ENDPOINTS

=====

POST

/auth/register

POST

/auth/login

POST

/auth/logout

POST

/auth/refresh

POST

/auth/forgot-password

POST

/auth/reset-password

GET

/auth/me

=====

=====

5. REGISTER

=====

=====

POST

/auth/register

REQUEST

email

password

first_name

last_name

RESPONSE

user

jwt

refresh_token

=====
=====

6. LOGIN

=====
=====

POST

/auth/login

REQUEST

email

password

RESPONSE

access_token

refresh_token

expires_in

user

=====

7. REFRESH TOKEN

=====

POST

/auth/refresh

REQUEST

refresh_token

RESPONSE

new_access_token

new_refresh_token

=====

8. CURRENT USER

=====

GET

/auth/me

RETURNS

Profile

Organization

Permissions

Preferences

=====

9. USER PROFILE API

=====

GET

/users/me

PATCH

/users/me

DELETE

/users/me

=====

10. USER SETTINGS

=====

GET

/users/me/preferences

PATCH

/users/me/preferences

=====

11. ORGANIZATIONS

=====

GET

/organizations

POST

/organizations

GET

/organizations/{id}

PATCH

/organizations/{id}

DELETE

/organizations/{id}

=====

12. ORGANIZATION MEMBERS

=====

GET

/organizations/{id}/members

POST

/organizations/{id}/members

PATCH

/organizations/{id}/members/{userId}

DELETE

/organizations/{id}/members/{userId}

=====

13. ROLE MANAGEMENT

=====

SUPPORTED

Owner

Admin

Architect

Editor

Viewer

=====

=====

14. AUTHORIZATION

=====

=====

JWT REQUIRED

Except

Register

Login

Forgot Password

Reset Password

=====

=====

15. COMMON STATUS CODES

=====

=====

200 OK

201 Created

204 No Content

400 Bad Request

401 Unauthorized

403 Forbidden

404 Not Found

409 Conflict

422 Validation Error

429 Too Many Requests

500 Internal Error

=====

=====

16. RATE LIMITS

=====

Login

10/min

Register

5/hour

API Default

120/min/user

=====

17. API VERSIONING

=====

URL VERSION

/api/v1

Future

/api/v2

=====

18. NEXT

=====

Projects API

Rooms API

Versions API

=====

END OF REST API SPECIFICATION PARTS 1 + 2

=====

AI HOME DESIGNER
REST API SPECIFICATION
PART 3 + PART 4

STATUS:
PROJECT API + ROOM API

=====

19. PROJECT ENDPOINTS

=====

GET

/projects

List user projects

POST

/projects

Create new project

GET

/projects/{projectId}

Get project details

PATCH

/projects/{projectId}

Update project

DELETE

/projects/{projectId}

Soft delete project

=====

20. PROJECT FILTERS

=====

SUPPORTED

status

owner

organization

project_type

tags

favorite

created_after

updated_after

search

Pagination

page

limit

sort

order

=====

21. CREATE PROJECT

=====

REQUEST

name

description

project_type

country

city

site_area

RESPONSE

project_id

current_version

status

=====

22. PROJECT DUPLICATION

=====

POST

/projects/{projectId}/duplicate

OPTIONS

Copy versions

Copy AI history

Copy exports

Copy permissions

=====

=====

23. PROJECT SHARING

=====

=====

POST

/projects/{projectId}/share

FIELDS

user_id

role

expiration_date

=====

=====

24. PROJECT FAVORITES

=====

=====

POST

/projects/{projectId}/favorite

DELETE

/projects/{projectId}/favorite

=====

=====

25. PROJECT VERSIONS

=====

=====

GET

/projects/{projectId}/versions

POST

/projects/{projectId}/versions

GET

/projects/{projectId}/versions/{versionId}

POST

/projects/{projectId}/versions/{versionId}/restore

=====

=====

26. ROOM ENDPOINTS

=====

=====

GET

/projects/{projectId}/rooms

POST

/projects/{projectId}/rooms

GET

/projects/{projectId}/rooms/{roomId}

PATCH

/projects/{projectId}/rooms/{roomId}

DELETE

/projects/{projectId}/rooms/{roomId}

=====

=====

27. ROOM CREATE REQUEST

=====

=====

FIELDS

room_type

room_name

floor_number

x

y

width

height

rotation

=====

=====

28. ROOM UPDATE

=====

=====

SUPPORTED

Resize

Move

Rotate

Rename

Change Type

Attributes

Materials

=====

=====

29. ROOM VALIDATION

=====

=====

SERVER CHECKS

- ✓ overlaps
- ✓ dimensions
- ✓ boundaries
- ✓ minimum size
- ✓ project ownership

=====

=====

30. ROOM SEARCH

=====

=====

FILTERS

room_type

floor

minimum_area

maximum_area

tag

=====

=====

31. ROOM COPY

=====

=====

POST

/rooms/{roomId}/duplicate

OPTIONS

Copy materials

Copy furniture

Copy openings

=====

32. BULK OPERATIONS

=====

POST

/rooms/bulk-update

POST

/rooms/bulk-delete

POST

/rooms/bulk-move

=====

33. PROJECT STATISTICS

=====

GET

/projects/{projectId}/statistics

RETURNS

Room count

Floor count

Area

Estimated Cost

AI Score

Energy Score

=====

34. PROJECT VALIDATION

=====

GET

/projects/{projectId}/validate

RETURNS

Errors

Warnings

Suggestions

Validation Score

=====

35. NEXT

=====

Walls API

Doors API

Windows API

Geometry API

=====

END OF REST API SPECIFICATION PARTS 3 + 4

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AI HOME DESIGNER
REST API SPECIFICATION
PART 5 + PART 6

STATUS:
WALL API + DOOR API + WINDOW API

=====

36. WALL ENDPOINTS

=====

GET

/projects/{projectId}/walls

POST

/projects/{projectId}/walls

GET

/projects/{projectId}/walls/{wallId}

PATCH

/projects/{projectId}/walls/{wallId}

DELETE

/projects/{projectId}/walls/{wallId}

=====

37. CREATE WALL

=====

REQUEST

room_id

start_x

start_y

end_x

end_y

thickness

height

wall_type

material_id

=====

38. WALL OPERATIONS

=====

SUPPORTED

Move

Resize

Split

Merge

Rotate

Change Material

Change Thickness

=====

39. WALL VALIDATION

=====

SERVER CHECKS

- ✓ Valid coordinates
- ✓ Length > 0
- ✓ Thickness > 0
- ✓ Connected geometry
- ✓ No duplicate walls

=====

40. DOOR ENDPOINTS

=====

GET

/projects/{projectId}/doors

POST

/projects/{projectId}/doors

GET

/projects/{projectId}/doors/{doorId}

PATCH

/projects/{projectId}/doors/{doorId}

DELETE

/projects/{projectId}/doors/{doorId}

=====

=====

41. CREATE DOOR

=====

=====

REQUEST

wall_id

room_from_id

room_to_id

door_type

width

height

position

opening_direction

hinge_side

=====

=====

42. DOOR VALIDATION

=====

=====

SERVER CHECKS

✓ Fits wall

✓ No overlap

✓ Valid dimensions

✓ Connected rooms

✓ Fire rating (optional)

=====

=====

43. WINDOW ENDPOINTS

=====

GET

/projects/{projectId}/windows

POST

/projects/{projectId}/windows

GET

/projects/{projectId}/windows/{windowId}

PATCH

/projects/{projectId}/windows/{windowId}

DELETE

/projects/{projectId}/windows/{windowId}

=====

44. CREATE WINDOW

=====

REQUEST

wall_id

room_id

window_type

width

height

sill_height

position

opening_type

glass_layers

=====
=====
45. WINDOW VALIDATION
=====
=====

SERVER CHECKS

- ✓ Inside wall
- ✓ No overlap
- ✓ Valid dimensions
- ✓ Glass layer supported
- ✓ Energy parameters valid

=====
=====
46. GEOMETRY API
=====
=====

POST

/geometry/validate

POST

/geometry/intersections

POST

/geometry/calculate-area

POST

/geometry/calculate-perimeter

=====
=====
47. GEOMETRY RESPONSE
=====
=====

RETURNS

Area

Perimeter

Volume

Intersection List

Geometry Errors

Warnings

=====
=====
48. BULK GEOMETRY
=====
=====

POST

/geometry/bulk-update

POST

/geometry/bulk-validate

POST

/geometry/optimize

=====

=====

49. PROJECT REBUILD

=====

=====

POST

/projects/{projectId}/rebuild

REGENERATES

Geometry Cache

Cost Cache

Statistics

Validation

=====

=====

50. NEXT

=====

=====

AI API

Cost API

Export API

Notifications API

=====

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END OF REST API SPECIFICATION PARTS 5 + 6

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AI HOME DESIGNER
REST API SPECIFICATION
PART 7 + PART 8

STATUS:

AI API + COST API + EXPORT API

=====

=====

51. AI GENERATION ENDPOINTS

=====

=====

POST

/ai/generate

POST

/ai/optimize

POST

/ai/validate

POST

/ai/explain

GET

/ai/history

=====

=====

52. AI GENERATE

=====

=====

REQUEST

project_id

project_version_id

prompt

house_type

floors

target_area

room_requirements

building_rules

budget_limit

RESPONSE

generation_id

layout_json

validation_score

estimated_cost

warnings

processing_time

=====
=====

53. AI OPTIMIZE

=====
=====

POST

/ai/optimize

SUPPORTED

Room placement

Circulation

Natural lighting

Energy efficiency

Construction cost

Material optimization

=====

54. AI VALIDATION

=====

RETURNS

Geometry Score

Building Rule Score

Accessibility Score

Energy Score

Construction Score

Overall Score

Each score

0–100

=====

55. AI HISTORY

=====

GET

/ai/history

FILTERS

project

user

date

model

status

=====

=====

56. COST API

=====

=====

GET

/projects/{projectId}/cost

POST

/projects/{projectId}/cost/recalculate

GET

/projects/{projectId}/cost/history

=====

=====

57. COST RESPONSE

=====

=====

RETURNS

Material Cost

Labor Cost

Equipment Cost

Transport Cost

Taxes

Contingency

Grand Total

Cost per m²

=====

58. MATERIAL PRICE UPDATE

=====

POST

/materials/update-prices

OPTIONS

Manual

Supplier Feed

CSV Import

API Import

=====

59. EXPORT API

=====

POST

/export/pdf

POST

/export/png

POST

/export/svg

POST

/export/json

POST

/export/dxf

POST

/export/ifc

=====
=====

60. EXPORT OPTIONS

=====
=====

SUPPORTED

2D Floor Plan

3D Model

Cost Report

Material List

Room Schedule

Construction Report

Full Project Package

=====
=====

61. EXPORT STATUS

=====

GET

/export/{exportId}

RETURNS

Queued

Processing

Completed

Failed

Download URL

=====

62. IMPORT API

=====

POST

/import/json

POST

/import/dxf

POST

/import/ifc

POST

/import/template

=====

=====

63. IMPORT VALIDATION

=====

=====

CHECKS

Schema

Geometry

Missing References

Duplicate Objects

Version Compatibility

=====

=====

64. NEXT

=====

=====

Notification API

Admin API

Webhook API

System API

Error Codes

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END OF REST API SPECIFICATION PARTS 7 + 8

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AI HOME DESIGNER
REST API SPECIFICATION
PART 9 + PART 10

STATUS:
NOTIFICATION API + ADMIN API + WEBHOOK API + SYSTEM API

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=====

65. NOTIFICATION API

=====

=====

GET

/notifications

GET

/notifications/{notificationId}

PATCH

/notifications/{notificationId}/read

PATCH

/notifications/read-all

DELETE

/notifications/{notificationId}

=====

=====

66. NOTIFICATION TYPES

=====

=====

AI_FINISHED

EXPORT_READY

PROJECT_SHARED

PROJECT_UPDATED

VERSION_CREATED

SYSTEM_ALERT

SECURITY_ALERT

BILLING

LICENSE

=====

=====

67. ADMIN USERS API

=====

=====

GET

/admin/users

GET

/admin/users/{userId}

PATCH

/admin/users/{userId}

DELETE

/admin/users/{userId}

POST

/admin/users/{userId}/suspend

POST

/admin/users/{userId}/activate

=====

=====

68. ADMIN PROJECT API

=====

=====

GET

/admin/projects

GET

/admin/projects/{projectId}

PATCH

/admin/projects/{projectId}

DELETE

/admin/projects/{projectId}

=====

=====

69. ADMIN SYSTEM API

=====

=====

GET

/admin/system/status

GET

/admin/system/health

GET

/admin/system/events

GET

/admin/system/statistics

=====
=====

70. SYSTEM HEALTH RESPONSE

=====
=====

RETURNS

Database Status

AI Provider Status

Cache Status

Storage Status

API Latency

Queue Size

Active Users

Uptime

=====
=====

71. WEBHOOK API

=====
=====

GET

/webhooks

POST

/webhooks

PATCH

/webhooks/{webhookId}

DELETE

/webhooks/{webhookId}

72. WEBHOOK EVENTS

SUPPORTED

project.created

project.updated

project.deleted

version.created

ai.completed

export.completed

user.created

user.updated

73. API KEYS

GET

/api-keys

POST

/api-keys

DELETE

/api-keys/{keyId}

PATCH

/api-keys/{keyId}

=====

=====

74. SYSTEM LOGS

=====

=====

GET

/system/logs

FILTERS

Date

Severity

Service

User

Project

=====

=====

75. STANDARD ERROR CODES

=====

=====

AUTH_INVALID

AUTH_EXPIRED

PROJECT_NOT_FOUND

ROOM_NOT_FOUND

VALIDATION_FAILED

GEOMETRY_INVALID

EXPORT_FAILED

IMPORT_FAILED

AI_TIMEOUT

AI_PROVIDER_ERROR

RATE_LIMIT_EXCEEDED

INTERNAL_ERROR

=====
=====

76. PAGINATION STANDARD

=====
=====

REQUEST

page

limit

sort

order

RESPONSE

total_items

total_pages

current_page

next_page

previous_page

=====

=====

77. API SECURITY

=====

=====

JWT REQUIRED

HTTPS ONLY

CORS

Rate Limiting

CSRF Protection

Input Validation

Output Sanitization

=====

=====

78. OPENAPI

=====

=====

GENERATED

OpenAPI 3.1

Swagger UI

Redoc

JSON Specification

=====

=====

79. API STATUS

=====

=====

REST API

100% SPECIFIED

Production Ready

Version

v1

=====
=====

80. NEXT DOCUMENT

=====
=====

FRONTEND SPECIFICATION

Contents

- Application Architecture
- UI Components
- Canvas Editor
- State Management
- Pages
- Design System
- Responsive Layout
- User Experience

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END OF REST API SPECIFICATION
REST API DESIGN COMPLETE

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